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CERTIFICATE

This is to certify that the project report entitled "GeoPhysicsLandslideNet: Physics-Closed Pentastream Architecture for Multimodal Landslide Segmentation" submitted by Samreedh Bhuyan, student of Kalinga Institute of Industrial Technology (KIIT), in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology, is a bona fide record of work carried out by him under the guidance of Dr. Anil Earnest at CSIR Fourth Paradigm Institute (CSIR-4PI), Bengaluru. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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A Project Report on

GeoPhysicsLandslideNet: Physics-Closed Pentastream Architecture for Multimodal Landslide Segmentation

Submitted in partial fulfilment of requirement for the award of the degree

Bachelor of Technology
Of
Kalinga Institute of Industrial Technology (KIIT)

By

Samreedh Bhuyan

Carried out under the guidance of
CSIR Fourth Paradigm Institute (CSIR-4PI)

Council of Scientific & Industrial Research
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Project Report

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Finally, I express my heartfelt gratitude to my parents, family members, and friends for their unwavering encouragement, patience, and moral support throughout the completion of this project.

I sincerely thank everyone who has contributed, directly or indirectly, to the successful completion of this work.

Samreedh Bhuyan

Project Report:
GeoPhysicsLandslideNet — Physics-Closed Pentastream Architecture for Multimodal Landslide Segmentation

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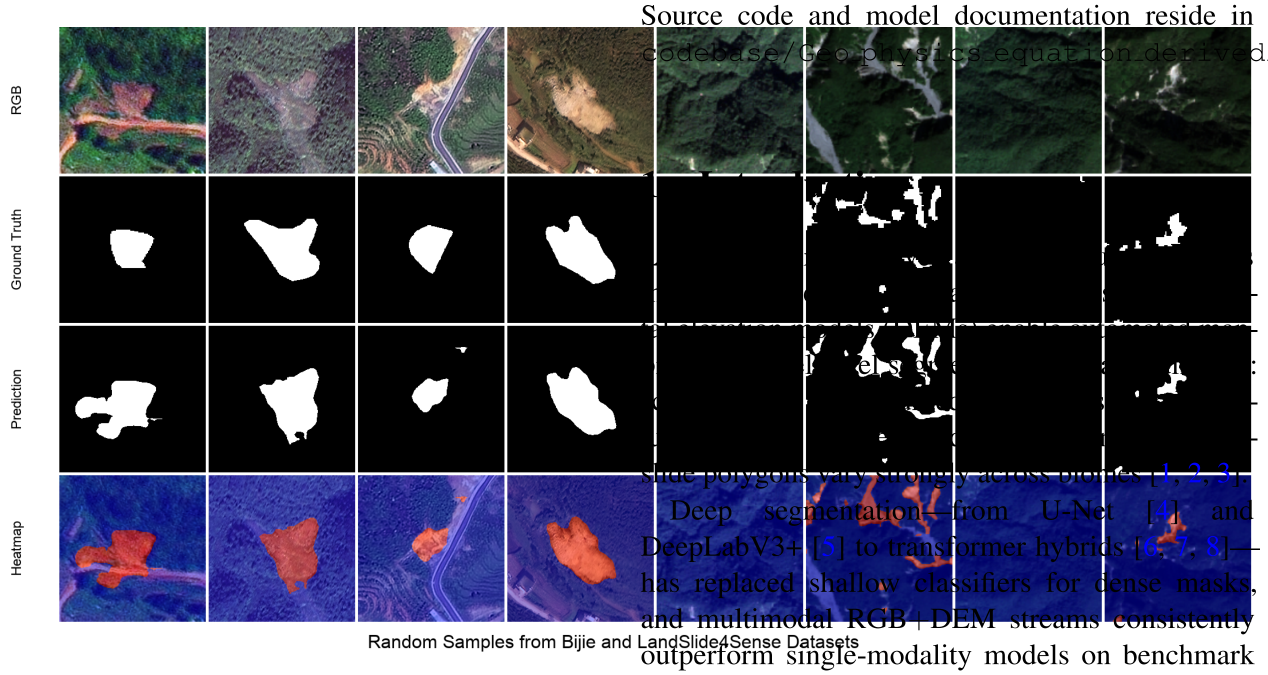

Figure 1: Opening qualitative panel (RGB, ground truth, prediction, heatmap) illustrating the landslide segmentation setting.

Abstract

This report presents the full research study on GeoPhysicsLandslideNet (PS-GPLNet), a pentastream physics-closed network for binary landslide segmentation from optical and topographic remote sensing inputs. The document includes the complete conference-style treatment: extended related work, mathematical formulation, implementation notes, quantitative and qualitative experiments on Bijie and Landslide4Sense, and discussion. The architecture embeds Taylor-stabilized infinite-slope factor-of-safety logic throughout encoding, fusion, decoding, and dual-path output equilibrium. Source code and model documentation reside in codebase/Geophysics_equation_derived_model/fr

slide polygons vary strongly across biomes [1, 2, 3].

Deep segmentation—from U-Net [4] and DeepLabV3+ [5] to transformer hybrids [6, 7, 8]—has replaced shallow classifiers for dense masks, and multimodal RGB+DEM streams consistently outperform single-modality models on benchmark tiles [3, 2, 9]. Dual-stream gated designs such as DiGATe-style networks learn to weigh spectral versus topographic evidence [10], but fusion and decoding remain statistically defined: gates are free parameters, not statements of slope stability.

Physics-informed geo-AI has advanced *susceptibility* mapping with infinite-slope factor of safety (FS), Newmark displacement, or hydromechanical priors in losses and sampling [11, 12, 13, 14, 15, 16, 17]. Pixel-accurate *segmentation* still seldom embeds the same mechanics as a first-class architectural primitive spanning encoding, fusion, and decoding. We therefore ask whether a single closed-loop FS inductive bias can organize a multimodal segmentation network end-to-end. PS-GPLNet is our answer: five parallel encoders hybridized by Complementary Modality Bridges (CMB), fused by MAO-GeoEGCA and TTEB, decoded by dual physics pathways with PGDI, and merged by Mechanistic Path Equilibrium Fusion (MPEF) driven by relative failure energy rather than scalar learned gates. Under a unified training protocol on Bijie [3] and Landslide4Sense [2], PS-GPLNet reaches best-epoch validation F1 = 0.933 / IoU = 0.907 on Bijie and F1 = 0.736 / IoU = 0.660 on Landslide4Sense, improving over strong CNN, transformer, and dual-stream baselines in our harness.

2 Related Work

2.1 From inventories to dense landslide mapping

Regional landslide studies historically combined field inventories with heuristic GIS overlays on slope, curvature, and hydrological indices [18, 19]. With open high-resolution optical stacks and DEMs, the community shifted toward *dense* polygon mapping: each pixel is labeled landslide or background, enabling direct comparison with building footprints, roads, and runout models. Bijie [3] and Landslide4Sense [2] crystallized this trend with public train/validation splits and reported CNN baselines. The task is imbalanced (scar pixels are rare), spectrally ambiguous, and topographically structured—properties that motivate both multimodal fusion and mechanistic priors.

2.2 Convolutional and transformer segmenters

Encoder–decoder CNNs remain the workhorse for geospatial segmentation. U-Net [4] and FCN [20] established skip-connected decoding; DeepLabV3+ [5] added atrous spatial pyramid pooling for multi-scale context; LinkNet and DEP-UNet variants in our ablation ladder inherit the same inductive bias. Transformers [21, 8] brought global context: Swin [22], SegFormer [6], and TransUNet [7] appear frequently in remote sensing leaderboards. These models excel at appearance and long-range dependencies but treat DEM derivatives as extra channels rather than as constraints from slope stability theory. PS-GPLNet retains EfficientNet [23] towers for representation learning while routing topographic proxies through parallel physics encoders.

2.3 Multimodal optical–topographic fusion

Fusing optical imagery with DEM, slope, aspect, or curvature is now standard for landslide detection [1, 3, 2]. Early fusion stacks channels at the input; mid-level fusion merges feature pyramids. BiFusionNet [9] and related multi-stream CNNs demonstrate that DEM-guided features reduce false positives on uniform steep slopes. Our Complementary Modality Bridge differs by *explicitly* pairing each CNN level with FS-gated physics features before tri-stream attention, so appearance and mechanistic maps are aligned per scale rather than only at the bottleneck.

2.4 Dual-stream and gated architectures

Dual-stream U-Nets separate RGB and DEM encoders and reunite them at the decoder [10]. DiGATe [10] adds gated attention between streams and an adaptive decoder. We include dual_stream_gated and Dual-Stream UNet as *empirical baselines only*. PS-GPLNet is not organized around learned scalar gates: path weights at the output arise from softmax failure energies (MPEF) computed from the same FS cell family used in the encoder inlet. Shared conventions—two geospatial inputs, deep supervision, Tversky or Dice losses [24, 25]—are field practice, not evidence of architectural equivalence.

2.5 Geotechnical susceptibility and slope stability

Infinite-slope FS compares resisting and driving forces on a potential slip surface [12]. Newmark [11] extends the framework to earthquake-induced displacements. These models underpin landslide susceptibility zoning when soil parameters are assigned from geology or calibrated statistically. They are interpretable and physically grounded but static: they do not, by themselves, produce segmentation masks at Sentinel or aerial resolution, nor do they ingest foundation-model embeddings. PS-GPLNet imports the FS *structure* as differentiable cells and gating signals, not as a post-classification threshold on a susceptibility raster.

2.6 Physics-informed and physics-guided learning

PINNs and physics-guided ML inject PDE residuals or FS inequalities into training [16, 13, 14, 15, 17]. Most geo-AI physics work targets susceptibility maps or point-wise failure probability, with physics entering through the loss or auxiliary heads. Differentiable rendering of geotechnical parameters has improved stability in recent susceptibility networks [13]. PS-GPLNet instead closes the loop *inside* the forward pass: bounded material parameters, pixel/latent mechanistic cells, PGDI skip injection, and MPEF routing all share one FS skeleton. Physics regularizes representation and routing, not only the final scalar objective.

Table 1: Conceptual positioning (dense segmentation setting).

Approach	Dense mask	FS in forward	Foundation context
CNN / transformer U-Net	✓	–	optional
Dual-stream gated [10]	✓	–	rare
Susceptibility PINN / PGML	often raster	loss / head	rare
PS-GPLNet (ours)	✓	✓	Prithvi+LoRA

2.7 Earth-observation foundation models

Large-scale pretraining on multi-temporal EO stacks—exemplified by Prithvi-EO [26, 27, 28]—provides transferable context for downstream segmentation and change detection. Full fine-tuning is costly; LoRA [29] adapts frozen backbones efficiently. We treat Prithvi as a fifth stream projected to the fusion width, then fuse with MAO-GeoEGCA against CMB-hybridized RGB/DEM features. To our knowledge, few landslide segmenters combine foundation-model context with dual mechanistic decode paths and FS-closed routing.

2.8 Positioning

Table 1 summarizes how PS-GPLNet relates to representative lines of work. The novelty claim is architectural closure: the same FS formalism at inlet, fusion, skips, decoder cells, and output equilibrium—implemented in `from_first_steps/` and documented in `model_architecture.md`.

3 Method

3.1 Problem and notation

Given streams $x_1 \in \mathbb{R}^{B\times 3\times H\times W}$ (RGB) and $x_2 \in \mathbb{R}^{B\times 3\times H\times W}$ (NDVI+slope+DEM on Landslide4Sense; replicated DEM on Bijie), predict a binary mask $y \in \{0,1\}^{B\times 1\times H\times W}$. Proxies (α, h, m)—slope angle, elevation/soil-column proxy, moisture/vegetation—are derived inside the forward pass.

The infinite-slope factor of safety is

$$\mathrm{FS} = \frac{c + \phi\, h \cos^2\alpha}{\gamma\, h \sin\alpha\cos\alpha + w_m m + \varepsilon}, \tag{1}$$

with positive material parameters obtained via bounded exponentials $c = \exp(\mathrm{clip}(w_c))$ (and likewise for ϕ, γ, w_m) for numerical stability. Failure energy $\mathrm{FE} = \mathrm{ReLU}(1 - \mathrm{FS})$ drives gating and path routing.

3.2 Pentastream encoding

Five encoders run in parallel: EfficientNet [23] RGB/DEM towers $E_{\mathrm{rgb}}, E_{\mathrm{dem}}$; PhysicsEncoders $P_{\mathrm{rgb}}, P_{\mathrm{dem}}$ with pixel/latent mechanistic cells; Prithvi+LoRA T_{fm}. At each EfficientNet level i, Complementary Modality Bridge (CMB) merges CNN features with spatially aligned physics features into hybrid streams $H^i_{\mathrm{rgb}}, H^i_{\mathrm{dem}}$.

3.3 Fusion: MAO-GeoEGCA and TTEB

At neck/bottleneck levels, MAO-GeoEGCA performs gated cross-attention between Prithvi and hybrid RGB/DEM maps. TTEB builds skip tensors by mixing neighbouring pyramid levels across three streams with a softplus stability residual $\delta = |\psi - R/(D+\varepsilon)|$.

3.4 Dual physics decoder and MPEF

Two PhysicsDecoders share fused context but use stream-specific (α, h, m) and bottleneck residuals. PGDI injects TTEB skips through a learned β gate on a latent mechanistic cell. Path logits are merged by MPEF:

$$[w_A, w_B] = \mathrm{softmax}([\mathrm{FE}_A, \mathrm{FE}_B]), \quad O = w_A O_A + w_B O_B + \mathrm{Co} \tag{2}$$

with entropy regularizer $\mathcal{R}_{\mathrm{MPEF}} = -\mathbb{E}[w_A \log w_A + w_B \log w_B]$.

Forward sketch. (1) Derive (α, h, m); encode five streams. (2) CMB hybrids; align Prithvi. (3) MAO at levels $2, 3$; TTEB skips at $0..3$. (4) Dual PhysicsDecoder $\to (O_A, O_B)$. (5) MPEF $\to$ (main, aux2, aux3, reg).

3.5 Closed-loop inductive bias

The pixel cell $g_{\mathrm{pix}}(x; \alpha, h, m) = f(x) \odot \sigma(\psi - \mathrm{FS}(\alpha, h, m))$ and MPEF weights $w = \mathrm{softmax}(\mathrm{FE})$ share the same FS skeleton, so proxies gate both features and path routing. Bounded $\exp(\mathrm{clip}(w))$ keeps FS finite during training.

3.6 Objective

Deeply supervised Tversky loss [24] on main/aux heads with weights $(1.0, 0.6, 0.4)$ plus $\lambda_{\mathrm{reg}} \sum_h \mathcal{R}^{(h)}_{\mathrm{MPEF}}$.

Table 2: Landslide4Sense best-epoch validation summary. PS-GPLNet (ours) at bottom.

Model	Acc	Prec	Rec	F1	IoU
LinkNet	0.973	0.487	0.735	0.431	0.359
DEP-UNet	0.977	0.669	0.770	0.603	0.535
GMNet	0.978	0.669	0.779	0.616	0.545
RMAU-Net	0.977	0.691	0.756	0.618	0.548
TransUNet	0.976	0.714	0.750	0.619	0.551
EMR-HRNet	0.975	0.674	0.784	0.627	0.557
ShapeFormer	0.975	0.675	0.802	0.632	0.560
Dual-Stream UNet	0.986	0.713	0.779	0.656	0.581
dual_stream_gated	0.984	0.822	0.673	0.646	0.573
U-Net	0.984	0.795	0.719	0.680	0.611
DeepLabV3+	0.984	0.789	0.749	0.710	0.636
PS-GPLNet (ours)	**0.986**	0.802	**0.788**	**0.736**	**0.660**

Table 3: Bijie best-epoch validation summary. PS-GPLNet (ours) at bottom.

Model	Acc	Prec	Rec	F1	IoU
DEP-UNet	0.982	0.850	0.934	0.831	0.799
RMAU-Net	0.981	0.859	0.924	0.830	0.798
ShapeFormer	0.984	0.892	0.899	0.840	0.810
TransUNet	0.983	0.889	0.908	0.845	0.815
EMR-HRNet	0.983	0.903	0.905	0.846	0.818
GMNet	0.983	0.900	0.909	0.855	0.825
U-Net	0.987	0.891	0.933	0.873	0.842
Dual-Stream UNet	0.986	0.902	0.928	0.875	0.843
dual_stream_gated	0.988	0.928	0.929	0.897	0.869
DeepLabV3+	0.989	0.929	0.966	0.927	0.899
PS-GPLNet (ours)	**0.991**	**0.956**	0.946	**0.933**	**0.907**

4 Experiments

4.1 Datasets and protocol

Bijie [3]: high-resolution optical+DEM landslide/non-landslide tiles. **Landslide4Sense** [2]: multi-region Sentinel patches with RGB–NDVI–slope–DEM. Inputs resized to 256×256. Optimizer Adam ($3 \cdot 10^{-4}$, weight decay 10^{-4}), Tversky α=0.3, β=0.7 on L4S and α=0.7, β=0.3 on Bijie, batch size up to 32 (Bijie batch 2 in our final run), 100 epochs, threshold 0.5. EfficientNet-B4 backbones frozen when pretrained; Prithvi frozen with LoRA. Metrics: accuracy, precision, recall, F1, IoU on the main head.

4.2 Baselines

We compare against LinkNet, DEP-UNet, GMNet, RMAU-Net, TransUNet, EMR-HRNet, ShapeFormer, Dual-Stream UNet, U-Net, DeepLabV3+, and dual_stream_gated under the same ablation harness. TriEncoderCFMNet is excluded (unpublished). PS-GPLNet numbers are best validation epoch from `outputs_final_bijie` and `outputs_final_l4s`.

4.3 Quantitative results

Table 2 and Table 3 report best-epoch validation scores. On Bijie, PS-GPLNet reaches F1 = 0.933 and IoU = 0.907, surpassing DeepLabV3+ (0.927/0.899) and dual_stream_gated (0.897/0.869). On Landslide4Sense, PS-GPLNet achieves F1 = 0.736 and IoU = 0.660, ahead of DeepLabV3+ (0.710/0.636) and the dual-stream variants in our ladder.

Figure 2 and Figure 3 visualize the F1/IoU spread across all harness models; the IoU ranking charts in Figure 4 and Figure 5 highlight that PS-GPLNet tops the L4S ladder and matches or exceeds the strongest CNN on Bijie. Heatmaps in Figure 6 and Figure 7 show precision–recall–F1–IoU jointly, where our model trades slightly lower recall on Bijie for higher precision relative to DeepLabV3+. Figure 8 and Figure 9 quantify gains over DeepLabV3+ at the best epoch.

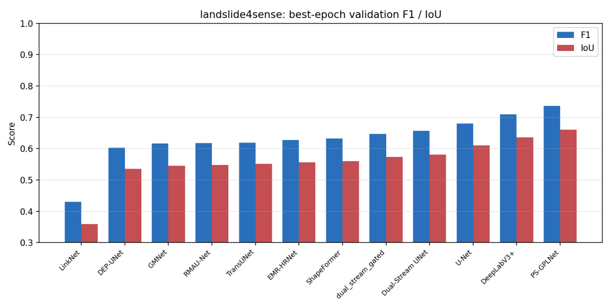

Figure 2: Landslide4Sense grouped metrics at each model's best validation epoch.

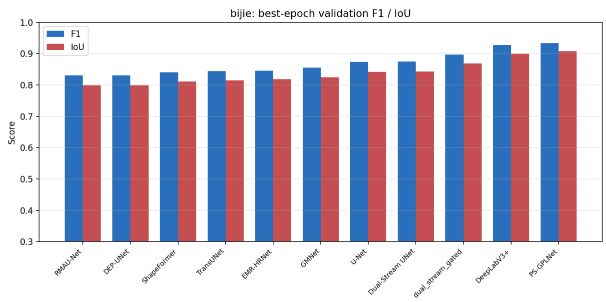

Figure 3: Bijie grouped metrics at each model's best validation epoch.

Training curves for PS-GPLNet are shown in Figure 10 and Figure 11. Precision–recall operating points for all models appear in Figure 12 and Figure 13.

4.4 Qualitative results

Representative Bijie and Landslide4Sense patches are shown in Figure 14 and Figure 15. On Bijie, poly-

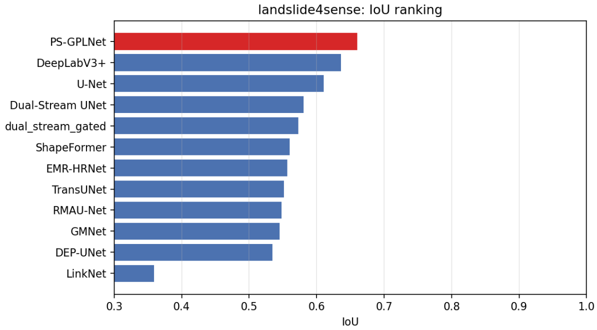

Figure 4: Landslide4Sense IoU ranking at best epoch.

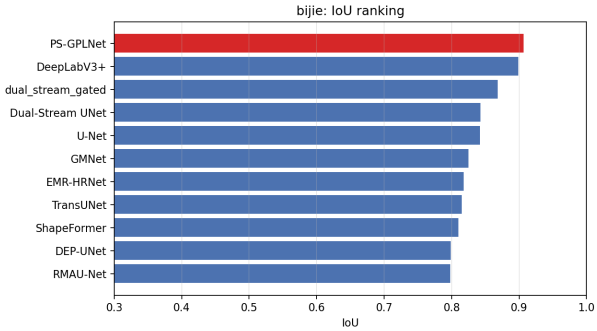

Figure 5: Bijie IoU ranking at best epoch.

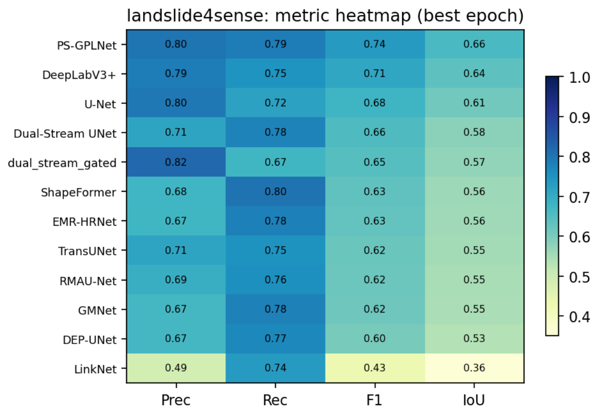

Figure 6: Landslide4Sense precision/recall/F1/IoU heatmap.

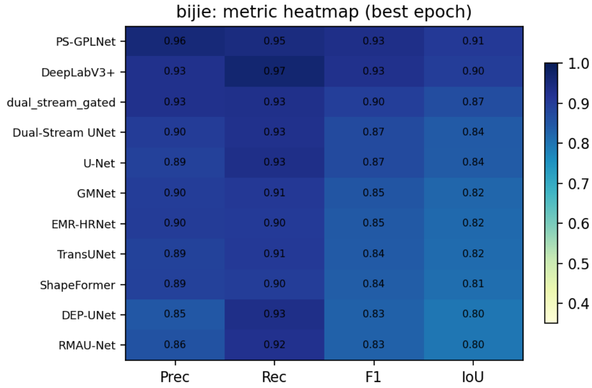

Figure 7: Bijie precision/recall/F1/IoU heatmap.

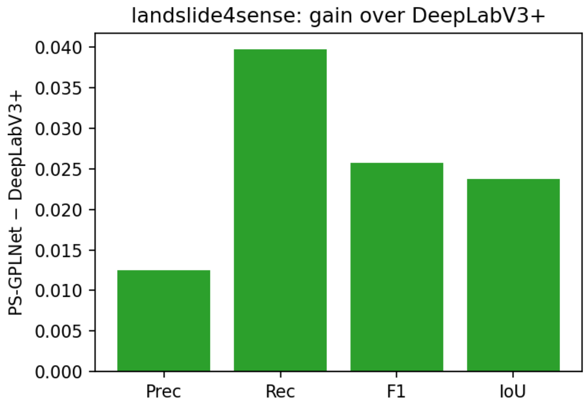

Figure 8: Landslide4Sense: PS-GPLNet minus DeepLabV3+ at best epoch.

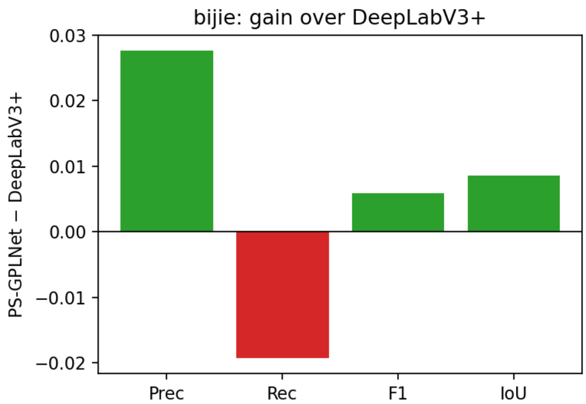

Figure 9: Bijie: PS-GPLNet minus DeepLabV3+ at best epoch.

gons are large and coherent; PS-GPLNet heatmaps concentrate on scarps and debris fans while suppressing homogeneous steep slopes. On Landslide4Sense, scars are smaller and more variable across regions; the model still recovers deposit extent but confuses some spectrally similar bare soil in shadowed valleys—consistent with the precision–recall trade-off in Table 2.

5 Discussion

Novelty. Unlike susceptibility PINNs [13], PS-GPLNet produces dense masks with foundation-model context. Unlike DiGATe [10], routing is failure-energy equilibrium (MPEF), and encoding is pentastream with CMB—not two EfficientNets plus scalar GateFuse.

Limitations. Full-width EfficientNet-B4 training requires sufficient GPU memory on shared HPC nodes. Compact presets reduce width and backbone size with-

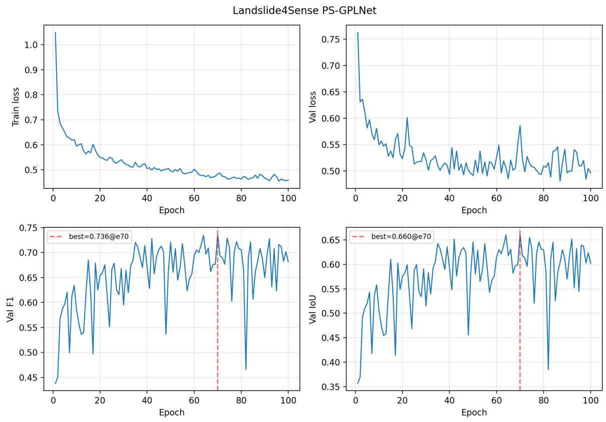

Figure 10: PS-GPLNet training dynamics on Landslide4Sense.

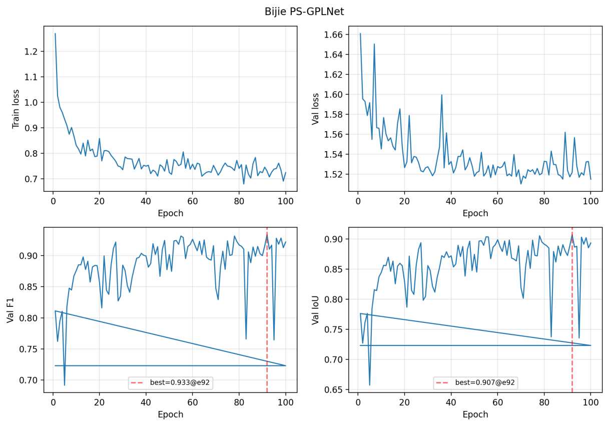

Figure 11: PS-GPLNet training dynamics on Bijie (best F1 at epoch 92).

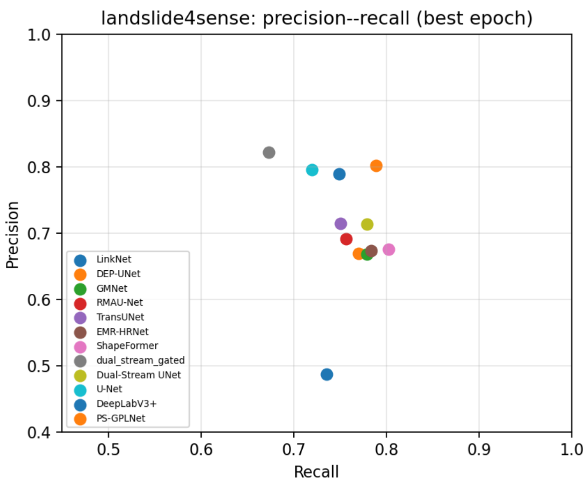

Figure 12: Precision–recall comparison on Landslide4Sense (best epoch per model).

out removing dual decoders. AUROC was not logged uniformly across baseline harness runs; we report thresholded F1/IoU for consistency.

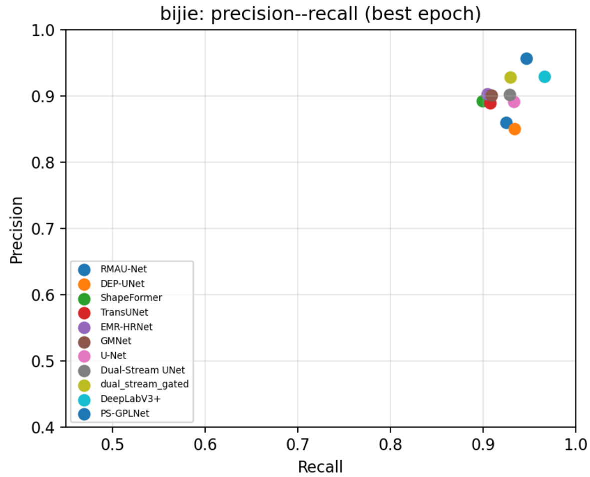

Figure 13: Precision–recall comparison on Bijie (best epoch per model).

6 Conclusion

We presented GeoPhysicsLandslideNet, a physics-closed pentastream landslide segmenter. Closing FS logic through encoding, fusion, decoding, and MPEF yields leading Bijie F1/IoU in our unified comparison and competitive Landslide4Sense scores. Future work includes ablating individual physics blocks, multi-temporal Prithvi conditioning, and coupling dynamic rainfall forcings into the moisture proxy m.

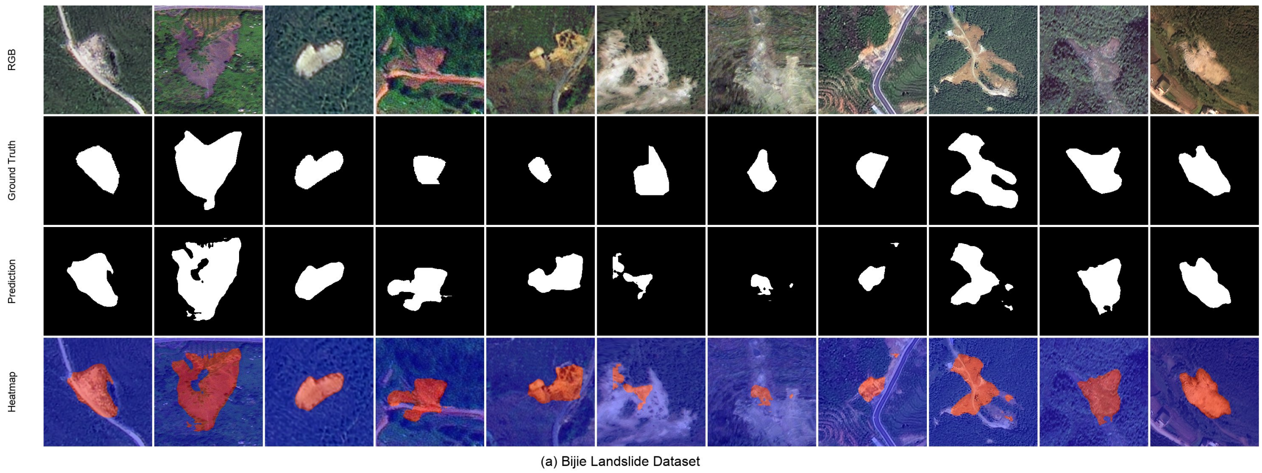

Figure 14: Qualitative results on Bijie validation samples: RGB, ground truth, prediction, and probability heatmap.

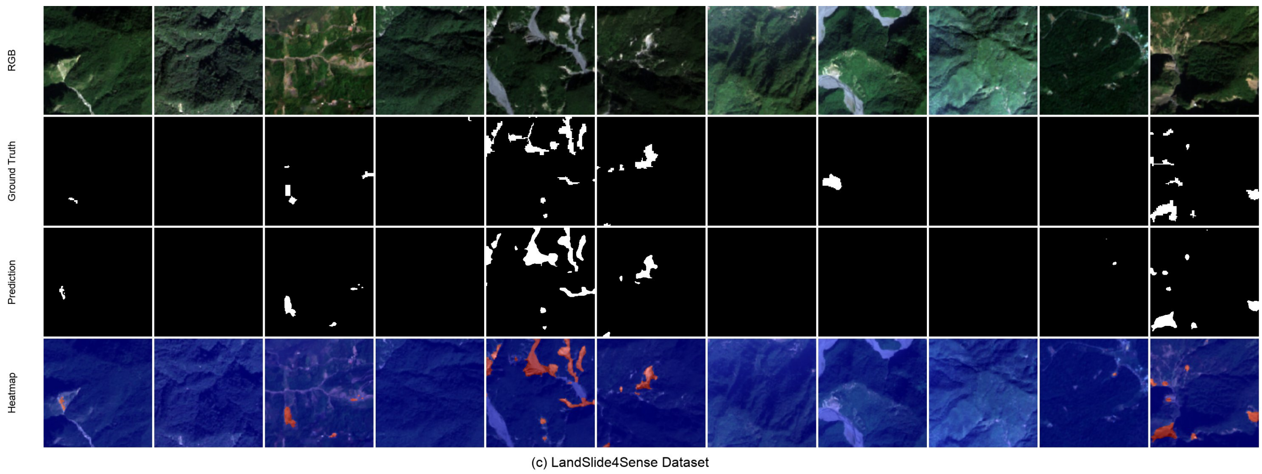

Figure 15: Qualitative results on Landslide4Sense validation samples.

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